Starter Flight Computer — Parts List

Sep 29, 2026 · @Alexander Agrawal

Parts list

Subsystem	Interface to MCU	Key parts
Power	ADC (battery voltage)	1S LiPo, FDN302P, AP2112K LDO
MCU core	—	STM32F405, 8 MHz crystal, reset/BOOT0 buttons
USB	USB	USB-C receptacle, USBLC6-2SC6
Debug and programming	SWD	2×5 header, STLINK-V3MINIE
Sensors	SPI1, I²C	LSM6DSO32, BMP585, MMC5983MA
GPS	UART	SAM-M10Q
Data logging	SPI2, SDIO	W25Q128JV, microSD socket
Buzzer	GPIO	CSS-J4B20 buzzer, AO3400A
Pyro	GPIO	2S connector, AO3400A ×3, AO3401A
Passives	—	All resistor and capacitor models

Power

The 1S cell powers all logic through a reverse-polarity FET and a 3.3 V LDO; the ADC reads battery voltage through the divider.

Part type	Part number
Logic battery (1S LiPo)	Adafruit 1578 3.7 V 500 mAh, JST-PH
Battery connector	JST S2B-PH-SM4-TB
Reverse-polarity FET	FDN302P P-channel

Part type	Part number
3.3 V LDO	AP2112K-3.3TRG1
Power LED	Lite-On LTST-C191KRKT red, 0603

MCU core

Everything the STM32F405 needs to boot and run: its clock, boot and reset controls, and a status LED.

Part type	Part number
MCU	STM32F405RGT6W (LQFP-64)
8 MHz crystal (HSE)	Abracon ABM3B-8.000MHZ-B2-T
Reset and BOOT0 buttons	C&K PTS810 SJM 250 SMTR LFS
Status LED	Lite-On LTST-C191KGKT green, 0603

USB

USB-C carries power, DFU flashing and log download.

Part type	Part number
USB-C receptacle	GCT USB4105-GF-A
USB ESD protection	USBLC6-2SC6

Debug and programming

Only the header goes on the board; the probe and cable are shared team tools.

Part type	Part number
SWD debug header	Samtec FTSH-105-01-F-DV-K 2×5 1.27 mm
Debug probe (off-board)	STLINK-V3MINIE
SWD cable (off-board)	Adafruit 1675 10-pin 1.27 mm

Sensors

The IMU and barometer share SPI1 with separate chip selects; the magnetometer sits alone on I²C.

Part type	Part number
IMU (accel + gyro)	LSM6DSO32TR
Barometer	BMP585
Magnetometer	MMC5983MA

GPS

The GPS talks to the MCU over UART and has its own lock LED.

Part type	Part number
GPS module	SAM-M10Q-00B (integrated patch antenna)
GPS lock LED	Lite-On LTST-C191KSKT yellow, 0603

Data logging

The microSD card (SDIO, 4-bit) is the main log; the SPI2 flash is the backup.

Part type	Part number
Flash (backup log)	W25Q128JVSIQ 16 MB SPI NOR
microSD socket	Molex 1040310811

Buzzer

A GPIO drives the magnetic buzzer through a low-side FET, with a flyback diode across the coil.

Part type	Part number
Buzzer	Same Sky CSS-J4B20-SMT-TR
Buzzer driver FET	AO3400A N-channel
Flyback diode	Diodes Inc 1N4148W-7-F

Pyro

A separate 2S pack fires the drogue and main channels through a high-side arm FET and one low-side fire FET per channel.

Part type	Part number
Pyro battery connector	JST S2B-XH-A
Pyro battery plug housing (off-board)	JST XHP-2
Pyro battery plug contacts (off-board)	JST SXH-001T-P0.6
Fire FETs (drogue, main, arm-gate driver)	AO3400A N-channel
Arm FET	AO3401A P-channel

Passives

Every resistor and capacitor on the board comes from this list. When your schematic needs one of these values, use this exact part so everyone builds from the same models.

Part type	Part number
Capacitor, 27 pF, 0603, C0G, 50 V	Samsung CL10C270JB8NNNC
Capacitor, 100 nF, 0603, X7R, 50 V	KEMET C0603C104K5RACTU
Capacitor, 1 µF, 0603, X5R, 50 V	Yageo CC0603KRX5R9BB105

Part type	Part number
Capacitor, 2.2 µF, 0603, X5R, 16 V	Samsung CL10A225KO8NNNC
Capacitor, 10 µF, 0805, X5R, 25 V	KYOCERA AVX KGM21AR51E106KU
Resistor, 1 kΩ, 0603, 1%	Yageo RC0603FR-071KL
Resistor, 4.7 kΩ, 0603, 1%	Yageo RC0603FR-074K7L
Resistor, 5.1 kΩ, 0603, 1%	Yageo AF0603FR-075K1L
Resistor, 10 kΩ, 0603, 1%	Yageo AR0603FR-0710KL
Resistor, 47 kΩ, 0603, 1%	Yageo RC0603FR-0747KL
Resistor, 100 kΩ, 0603, 1%	Yageo RC0603FR-07100KL